

Report 1 – Lamport Clock Analysis

Implemented rules

Local schedule: increment L, create the event, and store the resulting L in the immutable event record.

Send: increment L before each neighbor send. Left and right transmissions are distinct send events.

Receive: set $L = \max(\text{local } L, \text{received message } L) + 1$ before merging the received calendar.

Final Lamport values

Node	Final Lamport
0	285
1	263
2	266
3	269
4	272
5	275
6	278
7	281
8	284
9	284

Ten send/receive checks

Message	Sender	L(send)	Receiver	L before	L after	Verified
N01-M0001	1	2	0	0	3	YES
N01-M0002	1	3	2	0	4	YES
N04-M0001	4	2	3	0	3	YES
N04-M0002	4	3	5	0	4	YES
N09-M0001	9	2	8	0	3	YES
N09-M0002	9	3	0	3	4	YES
N06-M0001	6	2	5	4	5	YES
N06-M0002	6	3	7	0	4	YES
N00-M0001	0	6	9	3	7	YES
N00-M0002	0	7	1	3	8	YES

Three-node propagation chain

Time	Receiver	Sender	Message	New events
2026-09-13T14:07:33.601Z	0	1	N01-M0001	1
2026-09-13T14:07:33.604Z	2	1	N01-M0002	1
2026-09-13T14:07:47.440Z	3	4	N04-M0001	1

Observation

Every analyzed delivery must satisfy $L(\text{receive}) > L(\text{send})$. Lamport timestamps provide a consistent logical order, but different timestamps do not prove that two independent schedule events are causally related.

Conclusion

The observed logs demonstrate the required Lamport increment rules for local scheduling, separate neighbor sends, and receipt. The receive rule ensures the receiving logical time advances beyond the sender timestamp.